

Lab: Man-in-the-Middle (MITM) Attack using ARP Poisoning

Introduction

In this Lab you will learn how to detect a simple man-in-the-middle (MITM) scenario caused by ARP spoofing by analysing packet captures with Wireshark. You will not perform any network attacks, scanning or spoofing, you will only open and examine the provided capture files. The exercises are safe and designed for learners who are new to networking and packet analysis.

Files used in this lab

You will use two packet capture files only (you will find them inside the “exercise” folder on the Desktop):

1. normal_arp_traffic_200.pcap (you can find this at `/home/coder/Desktop/exercise/normal_arp_traffic_200.pcap`)
2. synthetic_mitm_arp_200_exact.pcap (you can find this at `/home/coder/Desktop/exercise/synthetic_mitm_arp_200_exact.pcap`)

Part A: Normal ARP behaviour

1. Open normal_arp_traffic_200.pcap in Wireshark.
2. Apply the display filter: arp
3. Click on any ARP packet and expand Address Resolution Protocol.
4. Observe that:
 - Each IP address maps to a single MAC address
 - The gateway IP address resolves to the same MAC address every time
 - There are no conflicting ARP replies

Part B: Identifying ARP poisoning

1. Open synthetic_mitm_arp_200_exact.pcap
2. Apply the display filter: arp.opcode == 2

Add columns:

- Ethernet II → Source → Apply as Column
- Address Resolution Protocol → Sender IP address → Apply as Column

Identify suspicious behaviour:

- The same IP address appears with different MAC addresses
- One MAC address appears repeatedly

Record:

- Attacker MAC address
- IP address being claimed
- Approximate number of suspicious ARP replies

Part C: Confirming MITM activity

1. Go to Statistics → Endpoints → Ethernet
2. Identify the MAC address that is neither gateway nor client but appears frequently

Verify traffic involvement.

Apply the filter:

eth.src == de:ad:be:ef:00:99 or eth.dst == de:ad:be:ef:00:99

Reflection:

Reflect on this exercise and what you have learn about the man in the middle attack.